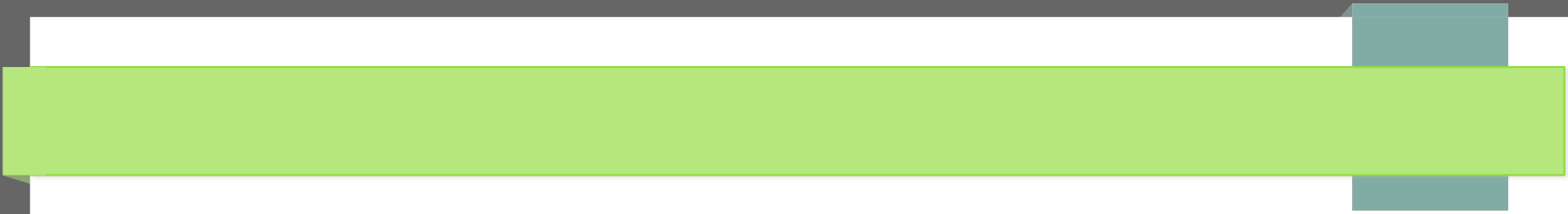


AI in HealthCare

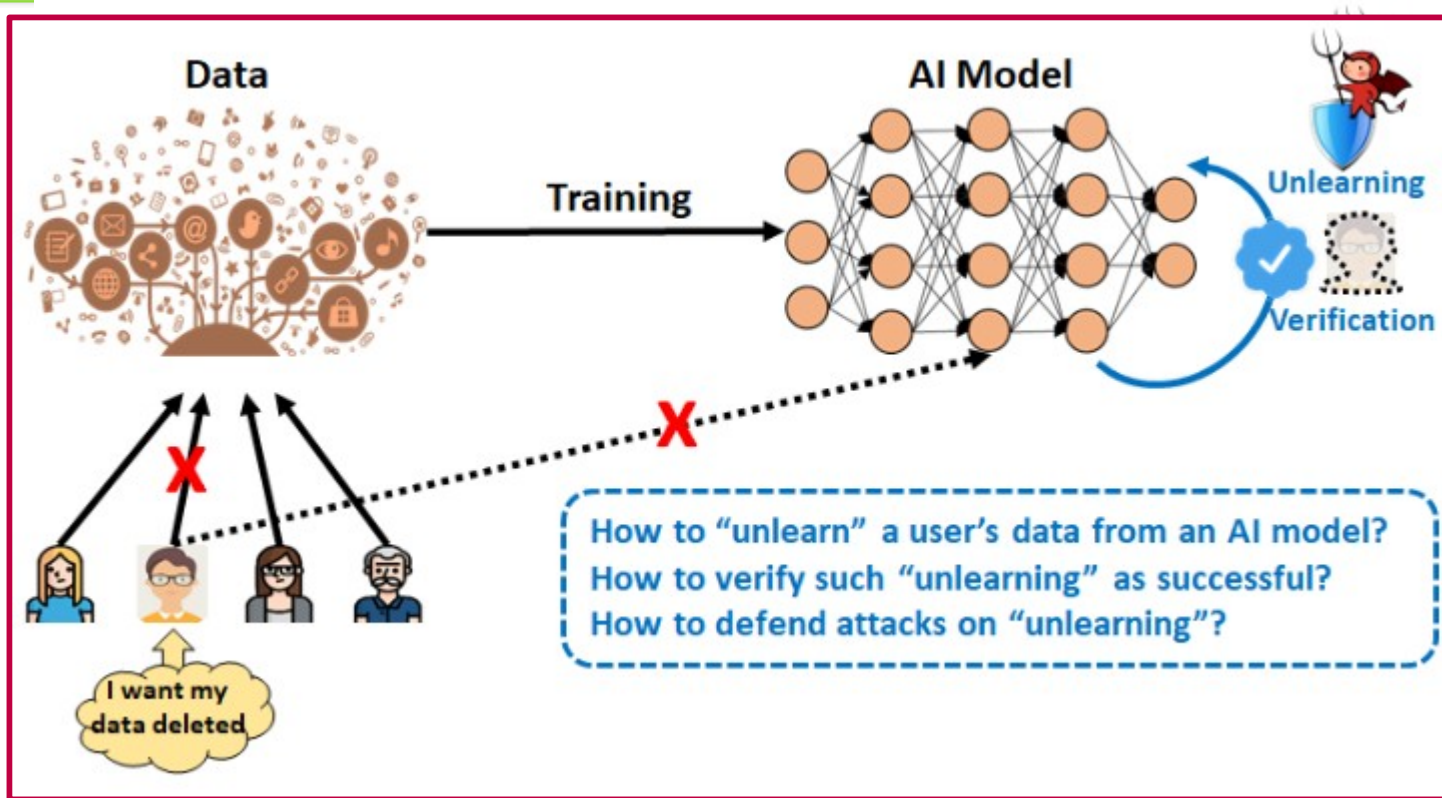
Prof. Swati Sharma

June 2025



Is it sufficient just to learn ??

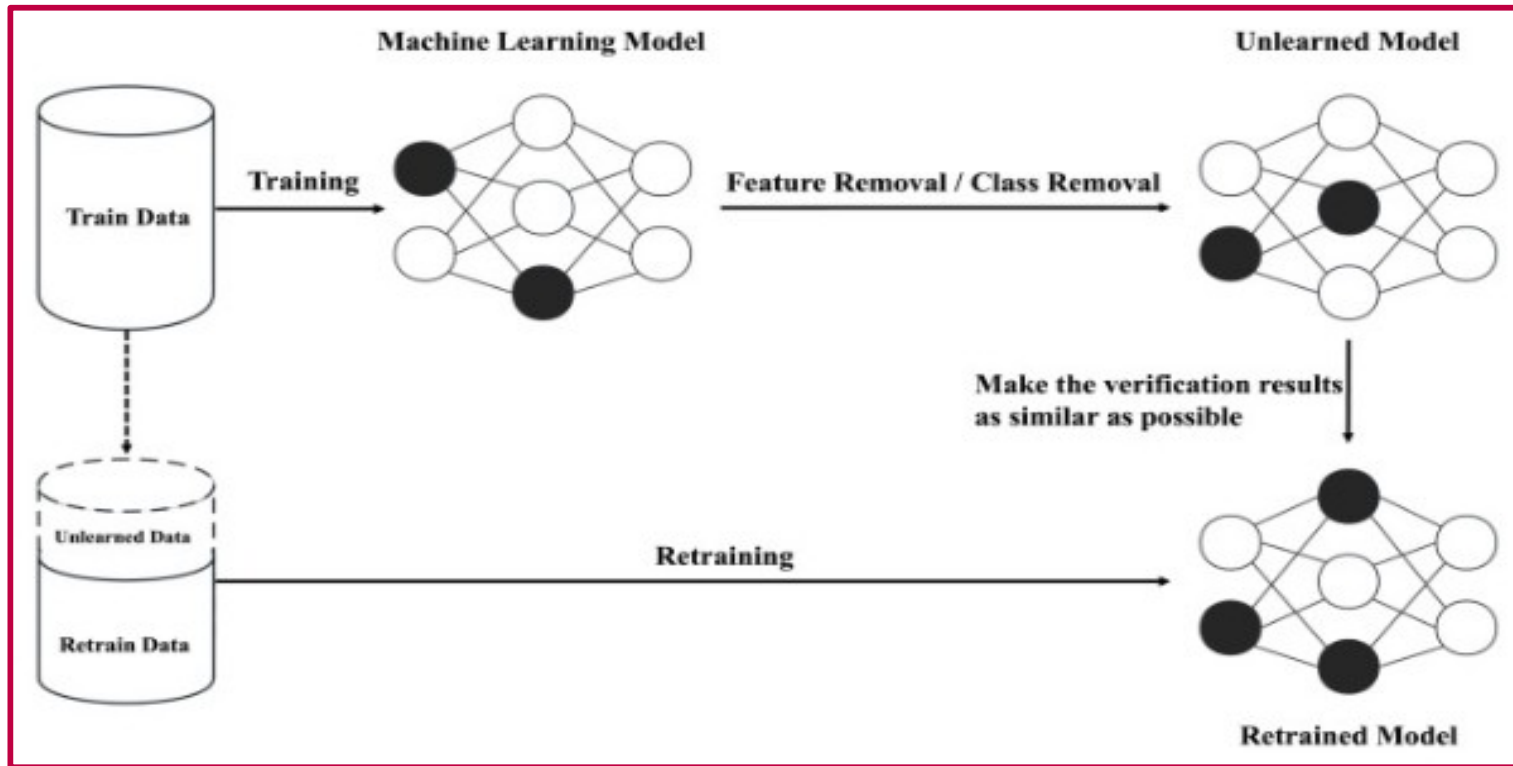
Machine Unlearning



Machine Unlearning

- ❖ Remove certain data points/features from a trained ML model.
- ❖ Without affecting its performance - sensitive/irrelevant/corrupt.
- ❖ Increasingly significant - fields - privacy, security, and fairness.
- ❖ Various institutions & governments - Argentina, European Union (EU), & Philippines - regulations around this issue.

Machine Unlearning



MU - Objectives

- ❖ Address privacy concerns - eliminating sensitive or personal data - without significantly reducing performance.
- ❖ Improve accuracy & fairness - removing biases or correcting errors introduced during learning.
- ❖ Improve performance over time - adapt to changing data & circumstances.
- ❖ Data Deletion - Data poisoning, data sub-sampling, & data shuffling.
- ❖ Data Perturbation - Data Anonymization, Differential Privacy (adding noise to data), Inverse Data Generation (generate a new dataset without any sensitive info!).
- ❖ Model Update -
 - ◆ Regularization (prevent overfitting),
 - ◆ Transfer Learning,
 - ◆ Distillation (size reduction for large & complex models while preserving accuracy),
 - ◆ Model Pruning (removing unnecessary parameters while maintaining performance),
 - ◆ Model Inversion (extract sensitive data from ML model - use a trained model to infer sensitive information about the data).

MU - Challenges

- ❖ Unlearn a single or multiple class(es) of data from a ML model without looking at the full training data even once?
- ❖ Make unlearning process fast and scalable to large datasets?
- ❖ Generalize it to different deep networks?

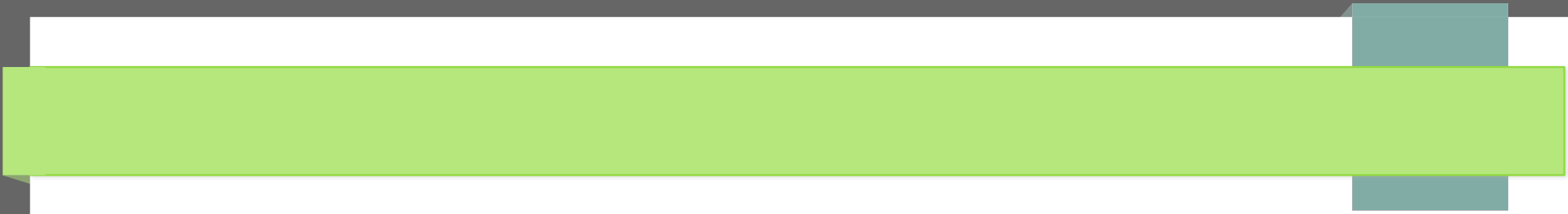
AI Ethics

❖ Bias-

- ◆ **Data Bias** - if data access – skewed/unrepresentative of population → biased outcomes.
- ◆ **Algorithmic Biases** – biased design choices in algo – discriminatory results.
- ◆ **Impact** – Biased AI can perpetuate societal inequalities – hiring, lending, criminal justice, etc.

❖ Security & Privacy -

- ◆ **Data Collection & Usage**- large personal data access – data collection, storage & usage.
- ◆ **Surveillance & Tracking** – Monitor individuals without their consent/knowledge.
- ◆ **Data Anonymization Limitations** – privacy risks in re-identification in certain cases.



That's all for Today !!